

CSE 340 Assignment 4

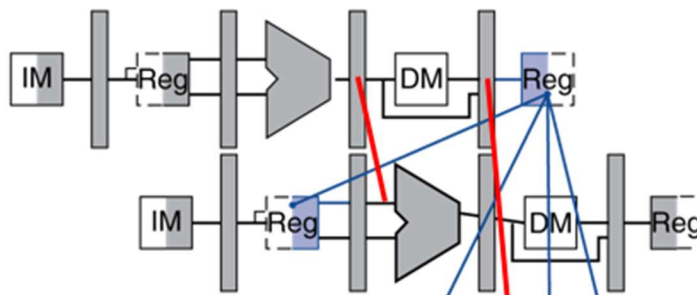
Deadline: 3 September 2026 11:30 PM

Submission link: <https://forms.gle/6xDD2zUfxVieo2yJ6>

Solve all five of the following questions:

1. Draw the full single cycle processor data path with control that can run R-type, load/store, beq instructions.
2. What is hazard? List different types of hazards that can occur in pipelining and mention ways to solve/reduce cpu stalls for each type of hazards.
3. Suppose while manufacturing a processor from silicon, in one unit, the ALUSrc Mux got stuck at 1 and MemtoReg got stuck at 0. Now list, which instructions will still be able to run correctly and which of them will fail to execute properly.
4. Consider this sequence:
sub \$2, \$1,\$3
and \$12,\$2,\$5
or \$13,\$6,\$2
add \$14,\$2,\$2
sw \$15,100(\$2)

Draw simplified pipeline time diagram of these instructions. Using a line show data dependencies between instructions. Also show where data needs to be forwarded between instructions. A sample diagram is given below for your reference.



5. Suppose opcode for all SB type instruction is same, (same opcode for all branch instruction). But func3 varies depending on the exact instruction. Meaning func3 for bge, blt, bne and beq are different. Now, Design a single cycle datapath that can accommodate beq, blt, bne instructions. Only draw the necessary components.